



Assignment
Lab Work 1

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Introduction

This report represents the fictional organization and project that our group will work with throughout the IT architecture course. The aim of Lab Work 1 is to create a realistic organization that can be used for later modelling tasks such as business modelling, BPMN, enterprise architecture and conceptual modelling amongst others.

Our organization is Nording Learning Solutions AB (NLS), a Swedish EdTech company offering a digital learning platform for schools, universities and possibly corporate training. The internal project we will focus on is the development of an AI-supported personalized learning module, which will analyze student data and provide individualized learning recommendations.

The structure of the organization and the project is based on ideas from the literature provided in Lecture 1, especially the concepts introduced by Sundgren, Langefors, Bubenko and Nilsson. Their views on information systems, information creation, modelling and organizational trends help us define NLS in a grounded and structured way.

1. Theoretical Background

Sundgren describes information systems as systems for concerted actions, meaning that people, processes and technology need to work together toward shared goals. This idea fits well with NLS, because the company's platform depends on coordinates activities between students, teachers, administrators and internal teams.

Langefors explains the information is created through a combination of data, the user's knowledge and the situation. This is highly relevant for an EdTech company. The AI module we plan to develop uses raw learner data and transforms it into meaningful recommendations, depending on the learner's context.

Bubenko emphasizes the importance of clear goals and stakeholder's needs before any system is designed. His modelling principles guide how we identify the requirements for NLS and the new AI module, making sure we understand the business environment before building anything.

Nilsson highlights long-term trends in information systems, such as increasing digitalization, integrated systems and user-centered design. NLS aligns with these trends by providing a flexible platform that supports modern learning and training needs.

Together, these theories give us a simple but strong foundation for defining our organization and its project.

2. Company Description

Nordic Learning Solutions AB is a mid-sized EdTech company founded in 2021 and based in Västerås. The company delivers a cloud-based learning platform used by educational institutions and companies for training and development.

NLS focuses on improving digital education by making learning more accessible, personalized and efficient. The platform allows teachers and trainers to create courses, monitor progress, design tests and give feedback. For students, the platform offers a clean interface for studying, completing exercises and receiving guidance.

The new AI module will become an extension of the platform and will help tailor learning content to individual needs.

3. Business Processes

NLS work through a series of core and support processes that define how the company delivers value.

Core processes

- Onboarding new customers
- Creating and publishing course materials
- Student registration and login
- Progress tracking
- Generating personalized recommendations (project focus)

Support processes

- GDPR and data protections processes
- System maintenance and updates

These processes will later be modeled using BPMN to make them clearer and easier to analyze.

4. Products and Services

The main product offered by NLS is its digital learning management platform. It includes tools for course creation, progress tracking, analytics and assessments. The platform also integrates with external tools to ease the learning curve in certain areas.

The AI-supported personalized learning module will be a new feature designed to analyze student interaction and recommend exercises, resources and learning paths. This follows Langefor's idea that information must be relevant and meaningful to the user's situation.

5. Stakeholders

The main stakeholders of NLS include:

Primary stakeholders

- Students, who use the platform for learning
- Teachers, who create content and track student progress
- School/Company administrators, who manage accounts and policies

Internal stakeholders

- Developers and IT architects, who build and maintain the system
- Data & Analytics team, responsible for the AI module
- Customer support, handling onboard and user issues
- Data Protection Officer, ensuring legal compliance

Each stakeholder has different goals and information needs, which fits Langefor's view that the meaning of information depends on the user's situation.

6. Organizational Structure

NLS is organized into several key departments:

- Executive Managements, responsible for strategy
- Product development, including backend, frontend and QA
- IT architecture & Infrastructure, ensuring technical stability
- Data & Analytics, developing the AI module and analyzing data
- Customer support

This structure supports cooperation across teams, reflecting Sundgren's idea that an information system is connected to how people work together.

7. Perspective and Success Factors

From the customer perspective, the platform should be easy to use, reliable and helpful in improving learning. Students want clear guidance, while teachers want tools that simplify their work.

From an employee perspective, team members benefit from well-documented and scalable technical environment.

For management, success depends on product quality, customer growth, financial sustainability and the ability to adapt to new market trends.

Key success factors include:

- High system reliability

- Scalability
- Strong alignment between business and IT
- User centered design
- Good data quality

These elements connect to Nilsson's trends in modern information systems.

8. Selected IT Architecture Frameworks

To analyze and model the organization, we will use several frameworks commonly taught in the course:

- Business Model Canvas (BMC) to describe value, customers and resources
- Balanced Scorecard (BSC) to link strategic goals to measurable indicators
- Business Architecture (BA) to connect processes, capabilities and goals
- BPMN to model workflows such as the personalized recommendation process

These frameworks help us follow Bubenko's principle of separating problems descriptions from technical implementation details.

Conclusion

This report has defined the fictional organization Nordic Learning Solutions AB and the internal project that our group will use throughout the IT architecture course. By applying ideas from Sundgren, Langefors, Bubenko and Nilsson, we have developed a clear and theoretically grounded description of the company, its goals, its stakeholders and its business processes.

References

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